ANUJ PUROHIT

FULL-STACK AI ENGINEER & AUTOMATION ARCHITECT

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer specializing in generative AI orchestration, Retrieval-Augmented Generation (RAG) architecture, agentic workflows, and scalable cloud automation platforms. Proven track record in building end-to-end full-stack web applications, high-throughput REST APIs, and database solutions using Python, FastAPI, React, TypeScript, and PostgreSQL. Demonstrates deep engineering versatility spanning modern web software, vector databases, machine learning workflows, and hardware-level STM32 microcontroller firmware development.

TECHNICAL SKILLS

AI & Generative Orchestration: LangChain, OpenAI API, Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone), n8n Workflow Automation, Agentic Workflows, OCR & ML Pipelines

Full-Stack Development: Python, FastAPI, Django, JavaScript (ES6+), TypeScript, React.js, Next.js, Node.js, Express.js, RESTful APIs, HTML5/CSS3

Databases & Cloud Infrastructure: PostgreSQL, Supabase, Redis, SQL Schema Design, Docker, Kubernetes, CI/CD Pipelines, AWS, Google Cloud Platform (GCP)

Embedded Systems & Firmware: STM32 Microcontrollers, Firmware Architecture, C/C++, Timers, PWM, Clock Peripherals, Hardware Debugging

WORK EXPERIENCE

STHERO — **Startup / Remote**

Full Stack AI Developer Intern — *2026*

- Architected and developed MVP1 of a full-stack platform for discovering, renting, and listing paying guest (PG) accommodations, taking ownership across product engineering, backend architecture, and database design.
- Engineered end-to-end user workflows for PG discovery, multi-parameter search filtering, real-time property listings, visit scheduling, and tenant-landlord communication channels.
- Designed and optimized normalized relational database schemas in **PostgreSQL** covering users, accommodations, visits, connection logs, and tenant reviews.
- Implemented secure authentication workflows utilizing **Supabase Auth** with granular role-based access controls (RBAC) separating tenant and landlord interfaces.
- Developed high-performance asynchronous backend microservices using **FastAPI** and integrated them seamlessly with a responsive **React/TypeScript** single-page application.

DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION (DRDO) — **Delhi, India**

Trainee — Embedded Systems and Firmware — *Trainee*

- Designed and programmed an **STM32-based** embedded control system focused on precise frequency stabilization for a laser-driven atomic clock.
- Developed low-level firmware in C/C++ leveraging micro-controller hardware timers, pulse-width modulation (PWM), and clock peripherals to support real-time signal stabilization.
- Optimized microcontroller performance, clock system configuration, and register-level execution for mission-critical hardware reliability.

DELHI UNIVERSITY — **Delhi, India**

Web & AI Intern — *Internship*

- Built a full-stack document processing application integrating **Optical Character Recognition (OCR)** and AI classification pipelines for automated document handling.
- Engineered RESTful backend APIs to extract, validate, and convert unstructured document formats into structured JSON data models for instant web application consumption.

FEATURED PROJECTS

FinPilot — AI-Powered Personal Finance Platform **React, TypeScript, FastAPI, PostgreSQL**

Full-Stack & AI Developer *Actively in Development*

- Architecting a full-stack personal finance platform enabling users to track expenditures, evaluate financial health metrics, and make data-driven budgeting decisions.
- Developing proprietary financial health scoring algorithms and AI-powered predictive models for monthly expense forecasting.
- Designing scalable REST APIs and optimized PostgreSQL data structures to store financial transaction records, user ledgers, and real-time analytics.
- Creating an intuitive, interactive dashboard presenting financial insights, spending velocity, and categorical expense breakdown charts.

TradeXpert — Stock Trading Simulator **React.js, Chart.js, PostgreSQL, FastAPI**

Lead Developer *Completed Project*

- Engineered a real-time stock trading simulator utilizing virtual currency, live market data feeds, and portfolio management tools.
- Integrated predictive AI models to evaluate price trends and explore short-term stock movement forecasts based on historical market indicators.
- Built high-performance dashboard visualizations using **Chart.js** and **React** to present simulated trading executions, equity growth curves, and market order books.

Sakhi — Multilingual AI Safety Agent **React.js, Chart.js, PostgreSQL, FastAPI**

Full-Stack AI Developer *Completed Project*

- Designed an AI assistant platform focused on women's education and safety featuring an anonymous storytelling and reporting interface.
- Implemented intelligent risk-detection analysis logic to evaluate user narratives and automatically flag concerning situations in real time.
- Developed dynamic legal recommendation engines helping users seamlessly explore relevant institutional, emergency, and legal support resources.

WorkSphere — AI SaaS Management Platform **React.js, Chart.js, PostgreSQL, FastAPI**

Full-Stack Engineer *Completed Project*

- Built an enterprise workforce management SaaS platform supporting employee onboarding, automated task allocation, and performance metrics tracking.
- Integrated AI-powered facial recognition verification and geofencing-based attendance logging workflows for automated workforce verification.
- Engineered robust backend microservices handling multi-tenant employee records, attendance history, and operational dashboard analytics.

EDUCATION

UNIVERSITY OF DELHI **Delhi, India**

Bachelor of Science in Electronics Science (Minor in Computer Science) *Expected Graduation: 2027*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Embedded Microcontrollers, Digital Signal Processing.